

Breast Cancer Diagnosis Prediction

The objective of this project was to develop and evaluate machine learning models capable of predicting whether a breast cancer tumor is malignant or benign based on clinical features. By exploring various classification algorithms, I aimed to understand how different computational approaches can assist in medical diagnostic tasks.

Dataset Description

The dataset used contains various quantitative measurements extracted from digitized images of fine needle aspirates (FNA) of breast masses. It includes features describing characteristics such as radius, texture, perimeter, and area of the cell nuclei. The target variable is the 'diagnosis', which classifies each sample as either malignant or benign. The data was pre-processed by removing irrelevant identifiers and missing values to ensure the training process was clean and robust.

Models Used

To determine the best approach for this diagnostic task, I trained three classical machine learning models:

Logistic Regression: A foundational statistical model often used for binary classification.

Random Forest: An ensemble method that uses multiple decision trees to improve predictive accuracy and control overfitting.

K-Nearest Neighbors (KNN): A proximity-based algorithm that classifies samples based on the characteristics of their neighbors in the feature space.

Following the initial training and evaluation, Logistic Regression demonstrated strong performance, achieving high accuracy and a reliable F1-score

Neural Network

To explore deep learning capabilities, I constructed a custom Neural Network. The architecture consisted of:

An input layer corresponding to the number of features in the dataset.

Two hidden dense layers (with 32 and 16 neurons respectively), utilizing ReLU activation functions to capture non-linear patterns.

A Dropout layer to mitigate overfitting and improve model generalization.

An output layer with a sigmoid activation function for binary classification.

The network was trained over 10 epochs using the Adam optimizer and binary cross-entropy loss function.

Results & Comparison

After evaluating all models, the Neural Network emerged as the top performer. It achieved the highest accuracy of approximately 98.25% and an F1-score of 98.13%. While the classical models (Logistic Regression, Random Forest, and KNN) provided excellent baselines, the Neural Network's ability to learn complex, non-linear relationships within the medical data allowed it to slightly outperform the traditional algorithms.

What I Learned

Through this project, I gained valuable insights into the machine learning workflow:

Data Pre-processing is Crucial: Scaling the data using `StandardScaler` was essential, particularly for algorithms like KNN and Neural Networks, which are sensitive to the magnitude of input features.

Model Complexity: While deeper models like the Neural Network can capture intricate patterns, classical models often provide competitive results with much lower computational costs.

Evaluation Metrics: Because the classes in medical datasets may not always be perfectly balanced, relying on the F1-score provided a more honest assessment of model performance than simple accuracy alone.